Anrui Wang

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EDUCATION

ShanghaiTech University, Shanghai, China Sep. 2023-Jul. 2027 (Expected)

- B.S. in Computer Science and Technology (AI Honors Program)
- Overall GPA: 3.58/4.00
- Relevant Coursework: Machine Learning, Deep Learning, Bioinformatics

RESEARCH EXPERIENCE

Deep Learning Framework for Inductive Protein–Protein Interaction (PPI) Prediction
Advisor: Prof. Xuming He (SIST, ShanghaiTech)
Dec. 2025-Sep. 2026

- To address the challenges of topological distortion in predicted PPI networks and the lack of network context for unseen proteins, we proposed GraSPPI, a motif graph prompt tuning framework for inductive PPI prediction.
- At the pair level, we designed a motif template based on triadic closure that infers each pair's shared interaction partners from proteins' intrinsic features and uses them as a prompt for the interaction decision.
- At the network level, we introduced structural losses that regularize the predictions on sampled training subgraphs jointly, so that the predicted network keeps the structure of the true network.
- Extensive experiments demonstrate that GraSPPI effectively preserves the topology of held-out PPI networks. It had the best average rank across multiple metrics among the compared methods.

Context-Specific Synthetic-Lethality (SL) Prediction with Pretrained Virtual Cell Foundation Models
Advisors: Prof. Jie Zheng (SIST, ShanghaiTech) and Prof. Lichun Jiang (SIAIS, ShanghaiTech)
Nov. 2025-Present

- Develop a deep learning framework based on pretrained AI Virtual Cell foundation models to predict how gene essentiality varies across different cancer cell lines, using multi-omics data.
- Extend the framework toward synthetic lethality (SL) discovery by using the predicted essentiality representations to predict context specific SL partners in held-out cell lines.
- Build a cell-line-specific SL prediction benchmark from multiple SL screening studies across diverse cell lines, and evaluate the model against multiple baselines and prior methods.

Deep Learning and Scientific Agent framework for Membrane PPI Prediction
Advisor: Prof. Xuming He (SIST, ShanghaiTech) and Prof. Wenqing Shui (iHuman Institute, ShanghaiTech)
Jun. 2025-Present

- Developed a deep learning model for endogenous transmembrane PPI prediction, including training algorithm design, model architecture optimization, and performance benchmarking.
- Developed an interactive research website to present our work and enable exploration of the tissue-wide transmembrane PPI network discovered through the study.
- Working on a combined wet-lab and dry-model development loop for membrane PPI prediction: the feedback from a small number of expensive wet-lab experiments lets an LLM scientific agent recursively improve its own workflow and the DL predictor, and the improved agent also guides the next wet-lab experiments.

PUBLICATIONS & MANUSCRIPTS

- **Wang, A.**, He, X. "**From Pairs to Networks: Motif Graph Prompt Tuning for Inductive Protein–Protein Interaction Prediction**." Manuscript under review at *the International Conference on Learning Representations **(ICLR)**, 2027.*
- Yang, L., Dang, T., Luo, H., Tan, Q., Li, A., Lang, X., Yang, X., Shi, S., Liu, X., **Wang, A.**, et al. "Proteomics-Informed Prediction of a Tissue-Wide Endogenous Transmembrane Protein Interaction Network." Manuscript under review at *Nature Biotechnology.*

SELECTED PROJECTS

CS290S Course Project: Official-Source RAG for ShanghaiTech/SIST QA May. 2026-Jun. 2026

- Optimized hybrid retrieval through BM25+dense retrieval, reciprocal-rank fusion, source diversity constraints, and reranking; improved expected-source hit@5 from 0.74 to 0.89 on a controlled benchmark
- Integrated a self-hosted Qwen generator with evidence-aware context selection, numbered citations, answer validation, and abstention mechanisms; increased correct answers in manual review from 59/100 to 69/100 and reduced abstentions from 18/100 to 14/100
- Conducted controlled ablation studies and failure analysis to evaluate the impact of indexing, fusion, reranking, and query-expansion strategies on retrieval and answer generation performance

CS181 Course Project: Hybrid Logic-DQN Minesweeper Solver Apr. 2025-Jun. 2025

- Designed a knowledge-based solver representing mine constraints as logical rules and resolving ambiguous states through exact enumeration and probabilistic sampling
- Designed a hierarchical policy that prioritizes guaranteed-safe moves, applies probability-guided actions under uncertainty, and integrates DQN as a fallback with multi-channel tensor state representations, experience replay, and target-network training
- Conducted controlled evaluations across constraint programming, probabilistic, pure-DQN, and hybrid strategies; achieved 80.90% win rate on 6×6 boards and 74.90% on 9×9 boards, outperforming baseline approaches

INTERNSHIP EXPERIENCE

Huida (Shanghai) Biotechnology Co., Ltd. Jul. 2025-Sep. 2025
AI Drug Data Retrieval Assistant R&D Intern — Shanghai, China

- Built a data and knowledge-base pipeline to process heterogeneous animal-study reports through schema parsing, cleaning, section extraction, semantic chunking, deduplication, and canonical normalization of dose, tumor volume, body/organ weight, and experimental timelines
- Implemented a retrieval and reasoning pipeline using LanceDB vector search, BM25 lexical scoring, weighted score fusion, metadata filtering, reranking, evidence-pack construction, query classification/decomposition, and section-level citation constraints to support traceable answers
- Integrated the pipeline into a bilingual ToxiRAG assistant with a Streamlit interface and command-line ingestion/evaluation tools to enable retrieval and extraction from previously unstructured toxicology reports
- Conducted benchmark-based evaluation and failure-case testing of the RAG system

AWARDS & COMPETITIONS

Creating a Moon Colony Using a Space Elevator System Jan. 2026-Feb. 2026
Mathematical Contest in Modeling (MCM) — Honorable Mention

- Modeled delivery of 100 million metric tons for a 100,000-person Moon colony, comparing space-elevator-only, rocket-only, and hybrid architectures across cost, timeline, reliability, water supply, and environmental impact
- Developed a time-dependent multi-commodity network-flow MILP in Pyomo, integrating transportation capacity, inventory constraints, costs, population growth, and lunar in-situ resource utilization
- Conducted comparative scenario experiments and 1,000-run Monte Carlo simulations with stochastic degradation and failure models to evaluate schedule, cost, and system robustness

NIPT Timing Optimization & Fetal Abnormality Determination Sep. 2025
Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM) — Second Prize, Shanghai Division

- Modeled Y-chromosome concentration with spline mixed-effects models accounting for repeated tests; compared Weibull/log-logistic accelerated failure-time models with Turnbull estimates to derive BMI testing windows
- Engineered chromosome Z-score, GC-content, and sequencing-quality features; compared Elastic-Net logistic regression and XGBoost with calibrated decision thresholds
- Evaluated model performance and robustness using patient-grouped cross-validation, held-out testing, PR-AUC/ROC-AUC, calibration, and Monte Carlo error propagation

SKILLS

Proficient in using AI agents.